

SAMARTH CHAUHAN *Software Engineer*

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EXPERIENCE

Backend and AI Intern Motorola Solutions	Jun 2025 – Present
- Built the cloud-based Agent Playground (shipped inside internal AI Foundry), letting developers and PMs connect live to MCP Servers, A2A-compatible Agents and configs in real time to inspect agent behavior. - Implemented isolated multi-user session environments (N users × M MCP servers) with per-session runtime using Python, Google ADK, and GCP Vertex AI SessionService — deployed across internal engineering and product teams. - Contributed to an enterprise RAG-based Google Chat agent (Node.js, AWS Bedrock, AWS S3); owned the real-time interaction layer, streaming live agent reasoning states (thinking, searching) to the client and rendering parsed markdown responses. - Developed an authorization middleware microservice in TypeScript securing internal AI Foundry infrastructure, handling complex OAuth 2.0 and Gated DCR OAuth flows with Redis session caching and MongoDB persistent state. - Evaluated frameworks Google ADK, CrewAI, AutoGen and architectures incorporating MCP and Agent Harness; findings directly informed the stack selection (ADK + Vertex AI SessionService) for the Agent Playground.	
Flutter Developer Intern Employee Forums	Nov 2024 – May 2025
- Architected and migrated a legacy Dart codebase to the BLoC pattern using comprehensive dependency injection. - Optimized cold startup time from 7s to 2s (71% improvement) by implementing lazy API invocation and deferred module loading. - Designed robust request/response data models and implemented SWR caching for seamless REST API integration.	

PROJECTS

VECSS - Distributed Video Transcoding System | *Golang, Python, Docker, RabbitMQ, FFmpeg* 

- Designed and implemented a scalable microservices-based platform for video transcoding using **Golang** and **Python**.
- Built an auto-scaling pipeline utilizing **RabbitMQ** and S3-compatible storage for asynchronous video processing.
- Containerized workloads with **Docker** and orchestrated **FFmpeg** workers to concurrently transcode videos into multiple resolutions.

Resume Cartel - Multi-Agent Resume Builder | *Python, Google ADK, Redis, Gemini* 

- Built a hierarchical multi-agent pipeline in **Python** using **Google ADK** — a root Coach agent that delegates to Writer, Critic, and YAML sub-agents, each reading and writing shared versioned markdown artifacts as inter-agent state.
- Backed agent memory with **Redis**; Agents write structured facts, notes, and file artifact references into a shared key-value store, giving sub-agents (Writer, Critic) read access to accumulated context without passing it through the LLM prompt each turn.
- Structured final output as typed YAML consumed by a downstream template engine, separating the LLM reasoning layer from document rendering and enabling the same data to drive multiple output formats.

Xginx - Layer 7 Load Balancer | *Golang* 

- Developed a custom **Layer 7** HTTP load balancer from scratch in **Golang**, engineering robust session-based routing and load distribution algorithms.
- Implemented automated, continuous health checks and retry mechanisms to actively monitor backend server pools, ensuring high availability and robust fault tolerance.
- Architected custom reverse proxy request handling to efficiently manage concurrent client connections.

SKILLS

Languages: Golang Python TypeScript(Node.js) Dart
AI & Agentic: Google ADK Model Context Protocol (MCP) Agent-to-Agent (A2A) AWS AgentCore GCP Vertex AI
Backend & Infra: Docker AWS S3 AWS Lambda GCP Cloud Run Redis MySQL RabbitMQ MongoDB
Frameworks: Gorilla/Mux FastAPI Flutter

EDUCATION

Charotar University of Science and Technology (CHARUSAT) B.Tech in Information Technology, CGPA 9.14	Anand, Gujarat Sep 2022 – Present
Shri Ramkrishna Harikrishna Academy Higher Secondary Certificate (CBSE), 90%	Surat, Gujarat Jan 2020 – May 2022

ACHIEVEMENTS

- Winner**, State Level Project Competition at Vidush Somany Institute.
- Winner**, SCIENCE Manthan Competition at P. D. Patel Institute.

RESPONSIBILITIES

Tech Lead CP Squad Club, CHARUSAT	Mar 2024 – Dec 2024
Orchestrated 5 technical contests with average participation of 400-500 students per contest, while managing competitive programming platforms.	